INFOSYS SE QUANT VAULT • PERCENTAGES, PROFIT & LOSS AND INTEREST

Successive Changes, CP/SP, Marked Price, SI, CI & Installments • Total Solved Questions: 50

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q1. A sum of Rs. 45500 is to be paid back in 3 equal annual installments. How much is each installment if the interest is compounded annually at 20% per annum ?

(A) 21600

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q2. X and Y enter into a business partnership with capital in the ratio 3:5. After 5 months, X adds 50% of his capital, while Y withdraws 60% of his capital. What is the share (in Rs lakhs) of X in the annual profit of Rs 10.26 lakhs?

(A) 5.89
(B) 4.56
(C) 5.58
(D) 6.24

Correct Answer: Option (C) : 5.58

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : 5.58.

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q3. A shopkeeper earns a profit of 20% by selling an article by giving 25% discount on its marked price. If the cost price of the article increases by 10%, then how much discount percent should he give now on the same marked price to earn the same percentage of profit as earlier?

(A) 17.5%
(B) 16.5%
(C) 18%
(D) 15%

Correct Answer: Option (A) : 17.5%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : 17.5%.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q4. A sum of money invested at compound interest amounts of 800 in 3 years and 882 in 5 years.What is the rate of Interest?

(A) 2.5%
(B) 4%
(C) ---
(D) 6.66%

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q5. The mark price of an article is Rs. 300.After allowing a discount 15% on the marked price, the shopkeeper makes a gain of Rs.15.What is the gain percentage?

(A) 16(1/4)
(B) 7(1/4)
(C) 16(2/3)
(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q6. Sum of CP's of two cows is Rs. 39, 000. Both the cows are sold at a profit of 20% and 40% respectively with their SP's being the same. What is the difference of CP's of both the cows?

(A) ---
(B) Rs. 2,000
(C) Rs. 1,500
(D) Rs. 2,500

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : ---.

[Campus Monk Day 03 Mock Test]
Topic: *Quantitative Aptitude*

Q7. Study the pie chart carefully and answer the questions. The pie chart shows the percentage breakup students in different classes from VI to X in the year 2017 1. The total number of students study in class VI and class IX together is approximately what percentage more than that of the total number of students study in class X?

(A) 125%
(B) 280%
(C) ---
(D) 189%

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 03 Mock Test]
Topic: *Quantitative Aptitude*

Q8. Study the pie chart carefully and answer the questions. The pie chart shows the percentage breakup students in different classes from VI to X in the year 2017 2. If in class VI, the ratio of boys to girls is 6:5 and the total number of girls in class VI is 50 less than that of the total number of girls in class VII. Find the ratio of the total number of boys in class VI to class VII.

(A) 5:7
(B) 7:11
(C) 9:13
(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : ---.

[Campus Monk Day 03 Mock Test]
Topic: *Quantitative Aptitude*

Q9. Study the pie chart carefully and answer the questions. The pie chart shows the percentage breakup students in different classes from VI to X in the year 2017 3. If the total number of boys in class VIII is "x" and the total number of boys in class IX is "x+40". The total number of girls in class VIII and class IX together is 306. Then find the value of "x".

(A) ---
(B) 210
(C) 174
(D) 184

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : ---.

[Campus Monk Day 03 Mock Test]

Topic: Quantitative Aptitude

Q10. Study the pie chart carefully and answer the questions. The pie chart shows the percentage breakup students in different classes from VI to X in the year 2017 4. What is the difference between the total number of students in class VII and class VIII together to that of the total number of students in class IX and class X together?

(A) 480
(B) 520
(C) ---
(D) 400

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 03 Mock Test]

Topic: Quantitative Aptitude

Q11. A man wants to invest Rs. 16850 in bank account of his 2 sons, whose ages are 12 years & 16 years in such a way so that they will get equal amount at an age of 120 years at the rate of 33(1/3)% compounded annually. Find the share of younger son?

(A) ---
(B) 4080
(C) 4090
(D) 4085

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : ---.

[Campus Monk Day 03 Mock Test]

Topic: Quantitative Aptitude

Q12. The simple interest (p.a.) accrued on an amount of Rs 17,000 at the end of four years is Rs 6,800. What would be the compound interest (compounded annually) accrued on the same amount at the same rate in two years?

(A) Cannot be determined
(B) Other than those given as option
(C) ---
(D) 3260

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 06 Mock Test]

Topic: Quantitative Aptitude

Q13. An almirah was sold at a profit of 15%. If its cost had been 5% less and it had been sold for Rs 1,470 less, then the profit would have been 10%. What is the cost price of the almirah?

(A) 14400
(B) 14000
(C) 16100
(D) 15400

Correct Answer: Option (B) : 14000

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : 14000.

[Campus Monk Day 06 Mock Test]

Topic: Quantitative Aptitude

Q14. Veer buys 200 guavas for Rs.1,800. Some of these guavas are rotten and are thrown away. He sells the remaining guavas at Rs.14 each and makes a profit of Rs.230. The percentage of guavas thrown away is:

(A) 27%
(B) 29%
(C) 30%
(D) 27.5%

Correct Answer: Option (D) : 27.5%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : 27.5%.

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q15. In an entrance exam of 200 objective questions, a student can score 1 point for every correct answer, loss 1/4 points for every wrong answer and loss 1/2 point for every unanswered question. If he attempts only 160 questions and he scores 100 points then the number of questions answered by him correctly is:

(A) A. 132
(B) A. 126
(C) A. 139
(D) A. 128

Correct Answer: Option (D) : A. 128

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : A. 128.

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q16. A good costs Rs 12000/-. A customer can choose either of the pairs of successive discounts. Pair A consists of successive discounts of 20% & 30% and pair B consists of 40% & 10%. Customer X chose A & customer Y choose B. What is the difference between their payments

(A) A) Rs 120
(B) A) Rs 240
(C) A) Rs 180
(D) A) Rs 60

Correct Answer: Option (B) : A) Rs 240

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : A) Rs 240.

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q17. A towel was 40 cm broad and 90 cm long. When bleached, it was found to have lost 20% of its length and 10% of its width. The percentage decrease in area is:

(A) 28%
(B) 30%
(C) 32%
(D) 33%

Correct Answer: Option (A) : 28%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : 28%.

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q18. Question : At 20% discount the selling price of a refrigerator is Rs 2400. What is the selling price if the discount is 32.5%?

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q19. Question : The length, Breadth and height of a cuboid increases by 20%, 10% and 30% Respectively. What is the percentage increases in the volume of that cuboid?

(A) 30%
(B) 20%
(C) 71.6%
(D) 43%

Correct Answer: Option (C) : 71.6%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : 71.6%.

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q20. The Line graph shows the data of 3 different types of cars sold in 5 different cities : Questio : Number of Honda car sold in Ahmedabad is what percent of the total Innova car sold in surat ?

(A) 50 %
(B) 66(2/3)%
(C) 70%
(D) None of these

Correct Answer: Option (B) : 66(2/3)%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : 66(2/3)%.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q21. Percentage of Marks obtained by various students in various subjects in an Examination. Q. The marks obtained by E in Geography are what percentage of the marks of the marks obtained by E in Hindi?

(A) 45
(B) 55
(C) > ■
(D) 60

Correct Answer: Option (C) : > ■

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : > ■.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q22. Percentage of Marks obtained by various students in various subjects in an Examination. Q. What are the average marks obtained by all the students together in science?

(A) 77.16
(B) 120.50
(C) 8GYWRhJk
(D) 121

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q23. A shopkeeper earns a profit of 20% by selling an article by giving 25% discount on its marked price. If the cost price of the article increases by 10%, then how much discount percent should he give now on the same marked price to earn the same percentage of profit as earlier?

(A) 17.5%
(B) 16.5%
(C) 18%
(D) 15%

Correct Answer: Option (A) : 17.5%

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : 17.5%.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q24. If A =x% of y and B =y%of x, then which of the following is true?

(A) Explanation: A=(x/100)
(B) A is smaller than B.
(C) A relationship between A and B cannot be determined.
(D) If x is greater than y, then A is less than B.

Correct Answer: Option (A) : Explanation: A=(x/100)

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : Explanation: A=(x/100).

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q25. A vendor bought mangoes at 6 for a rupee. How many for a rupee must he sell to gain 20%?

(A) 3
(B) 6
(C) 4
(D) Explanation: cost price of 1 mango =1/6 gain =20%=1/5 the selling price of 1mango =1/6+1/6

Correct Answer: Option (D) : Explanation: cost price of 1 mango =1/6 gain =20%=1/5 the selling price of 1mango =1/6+1/6

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : Explanation: cost price of 1 mango =1/6 gain =20%=1/5 the selling price of 1mango =1/6+1/6.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q26. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 :7 :8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

(A) Originally, let the number of seats for Mathematics, Physics and Biology be 5x, 7x and 8x respectively. Number of increased seats are (140% of 5x), (150% of 7x) and (175% of 8x).[140/(100
(B) 6 : 7 : 8
(C) 6 : 8 : 9
(D) 7 : 6 : 9

Correct Answer: Option (A) : Originally, let the number of seats for Mathematics, Physics and Biology be 5x, 7x and 8x respectively. Number of increased seats are (140% of 5x), (150% of 7x) and (175% of 8x).[140/(100

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : Originally, let the number of seats for Mathematics, Physics and Biology be 5x, 7x and 8x respectively. Number of increased seats are (140% of 5x), (150% of 7x) and (175% of 8x).[140/(100.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q27. If A + B means A is the sister of B; A x B means A is the wife of B, A % B means A is the father of B and A - B means A is the brother of B. Which of the following means T is the daughter of P?

(A) P x Q % R + S – T
(B) P x Q % R - T + S
(C) P x Q % R + T – S
(D) P x Q % R + S + T

Correct Answer: Option (B) : P x Q % R - T + S

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : P x Q % R - T + S.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q28. Statement : In a one day cricket match, the total runs made by a team were 200. Out of these 160 runs were made by spinners. Conclusions : I. 80% of the team consists of spinners. II. The opening batsman were spinners.

(A) Only I is implicit
(B) Only II is implicit
(C) Both I and II are implicit
(D) Neither I nor II are implicit

Correct Answer: Option (D) : Neither I nor II are implicit

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : Neither I nor II are implicit.

[Campus Monk Day 09 Mock Test]
Topic: Quantitative Aptitude

Q29. Statement: A major retail store announced thirty percent reduction on all food items during the weekend. Assumption: I. People may still prefer buying food items from other stores. II. A large number of customers may visit the retail store and buy food items.

(A) Only I is implicit
(B) Only II is implicit
(C) Both I and II are implicit
(D) Neither I nor II are implicit

Correct Answer: Option (B) : Only II is implicit

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : Only II is implicit.

[College ITM Mock Paper 1]
Topic: Mathematical Critical Thinking

Q30. A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:

(A) 7.1%
(B) 5.5%
(C) 8.6%
(D) 6.4%

Correct Answer: Option (A) : 7.1%

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]
Topic: Mathematical Critical Thinking

Q31. In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?

(A) 1520
(B) 1600
(C) 1360
(D) 1440

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]
Topic: Mathematical Critical Thinking

Q32. The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ■310. Find the sum.

(A) ■8,000
(B) ■9,000
(C) ■10,000
(D) ■12,000

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]
Topic: Mathematical Critical Thinking

Q33. ■12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ■2,270. How much was invested at simple interest?

(A) ■4,000
(B) ■5,000
(C) ■6,000
(D) ■7,000

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking

Q34. A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.

(A) 20%
(B) 22.2%
(C) 25%
(D) 18%

Correct Answer: Option (A) : 20%

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking

Q35. A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?

(A) 27,000
(B) 26,500
(C) 26,000
(D) 27,500

Correct Answer: Option (A) : 27,000

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking

Q36. A sum of money amounts to ■13,000 after 3 years and to ■15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.

(A) 8%
(B) 10%
(C) 12%
(D) 9%

Correct Answer: Option (A) : 8%

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking

Q37. ■16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ■18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).

(A) 9.5%
(B) 10.5%
(C) 11.5%
(D) 12.5%

Correct Answer: See Logic

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking

Q38. An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.

(A) 12 kg
(B) 10 kg
(C) 15 kg
(D) 8 kg

Correct Answer: Option (A) : 12 kg

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q39. ■8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.
(A) ■20
(B) ■40
(C) ■80
(D) ■100
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q40. In what time will ■8,000 amount to ■9,261 at 5% per annum, compounded annually?
(A) 2 years
(B) 3 years
(C) 4 years
(D) 3.5 years
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q41. A trader buys 80 kg of rice at ■25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?
(A) ■26.67
(B) ■25
(C) ■28
(D) ■24
Correct Answer: Option (A) : ■26.67
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q42. In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?
(A) 30,000
(B) 28,000
(C) 32,000
(D) 24,000
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q43. Find the compound interest on ■12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).
(A) ■3,860
(B) ■3,800
(C) ■3,960
(D) ■3,720
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q44. The compound interest on a sum for 2 years is ■1,050, and for 3 years is ■1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ■1,000. Find the rate of interest per annum.
(A) 10%
(B) 12%
(C) 8%
(D) 15%
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q45. A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ■200 (costing him ■120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?
(A) 25%
(B) 20%
(C) 30%
(D) 15%
Correct Answer: Option (A) : 25%
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q46. At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?
(A) 15 years
(B) 20 years
(C) 10 years
(D) 25 years
Correct Answer: Option (A) : 15 years
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q47. ■20,000 is borrowed at 10% per annum compound interest (annual compounding). ■11,000 is repaid at the end of year 1. What amount must be paid at the end of year 2 to clear the entire debt?
(A) ■12,100
(B) ■12,000
(C) ■11,500
(D) ■13,200
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]
Topic: Mathematical Critical Thinking
Q48. A merchant sells two articles at ■1,200 each. On one he gains 25%, on the other he loses 20%. What is his overall result (approximately)?
(A) 2.4% loss
(B) 2.4% profit
(C) No gain, no loss
(D) 5% loss
Correct Answer: Option (A) : 2.4% loss
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: Mathematical Critical Thinking

Q49. A man spends 40% of his salary on food, 25% of the remainder on rent, and 20% of what is then left on travel. He is left with ■3,600. What is his salary?

(A) ■10,000
(B) ■12,000
(C) ■9,000
(D) ■11,000

Correct Answer: Option (A) : ■10,000

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: Mathematical Critical Thinking

Q50. A loan is repaid in two equal annual installments of ■2,420 each, with interest at 10% per annum compounded annually (first installment due at the end of year 1). What was the loan amount?

(A) ■4,200
(B) ■4,400
(C) ■4,000
(D) ■4,840

Correct Answer: Option (A) : ■4,200

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.